



Tool Documentation

INDUSTRIAL ENERGY EFFICIENCY AND PRODUCTIVITY TOOLKIT

USER GUIDE FOR LEVELIZED COST CURVE TOOL

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Introduction to the Tool

Simple payback period (SPP) has been the standard metric to communicate techno-economics of energy efficiency measures for years. This has resulted in the familiarity of industrial executives and operators with the term to assess the economic viability of the recommended measures. Unfortunately, due to the nuances with electrification measures, it may not be appropriate to use simple payback period for comparing the assessment recommendations for measures like industrial heat pumps, for example. Such measures do not always yield net savings, resulting in a negative payback period confusing the decision makers at a facility. Therefore, a more appropriate and intuitive metric for comparing the assessment recommendations (ARs) is needed. Metrics like the Levelized Cost of Conserved Energy (LCCE) use the lifetime costs and savings associated with a measure and divide it by the amount of conserved energy to compare a broader range of measures than SPP. An intuitive bar chart that plots this cost for multiple assessment recommendations on the vertical axis and adjusts the bar width for each measure to represent the energy it conserves is similar to the popular Marginal Abatement Curve (MAC). This curve communicates the marginal cost of abating the next unit of energy at every level of progress towards the highest efficiency at a facility. It gained popularity in late 90s among environmental researchers, policymakers, and management consultancy firms for sector-level analyses. Bringing this high-level visual comparison graphic to small-to-medium industries not only helps them better understand the techno-economics of a wider range of ARs but also makes the decision-making more robust.

For industries, this levelized cost curve communicates the cost of conserved energy in \$/MMBtu to get to the total efficiency improvement resulting from the assessment recommendations. This curve, as generated by this tool, is demonstrated as an example in Figure 1 where about 35% of the total energy conservation comes from net-savings measures like efficiency and onsite solar. Industrial heat pumps and electric boilers are capable of conserving about 65% of the total achievable conservation as per this example assessment report. The horizontal dashed line indicates the weighted average levelized cost of conserved energy for this facility which is a net-cost of \$2/MMBtu. The chart also has other important bits of information like the annualized cost of implementing this set of recommendations based on the facility's discount rate and the total of conserved energy. The intuitiveness and information density of this figure is well suited to an industrial assessment. Unlike the simple payback period, understanding the negative and positive levelized costs is much more intuitive.

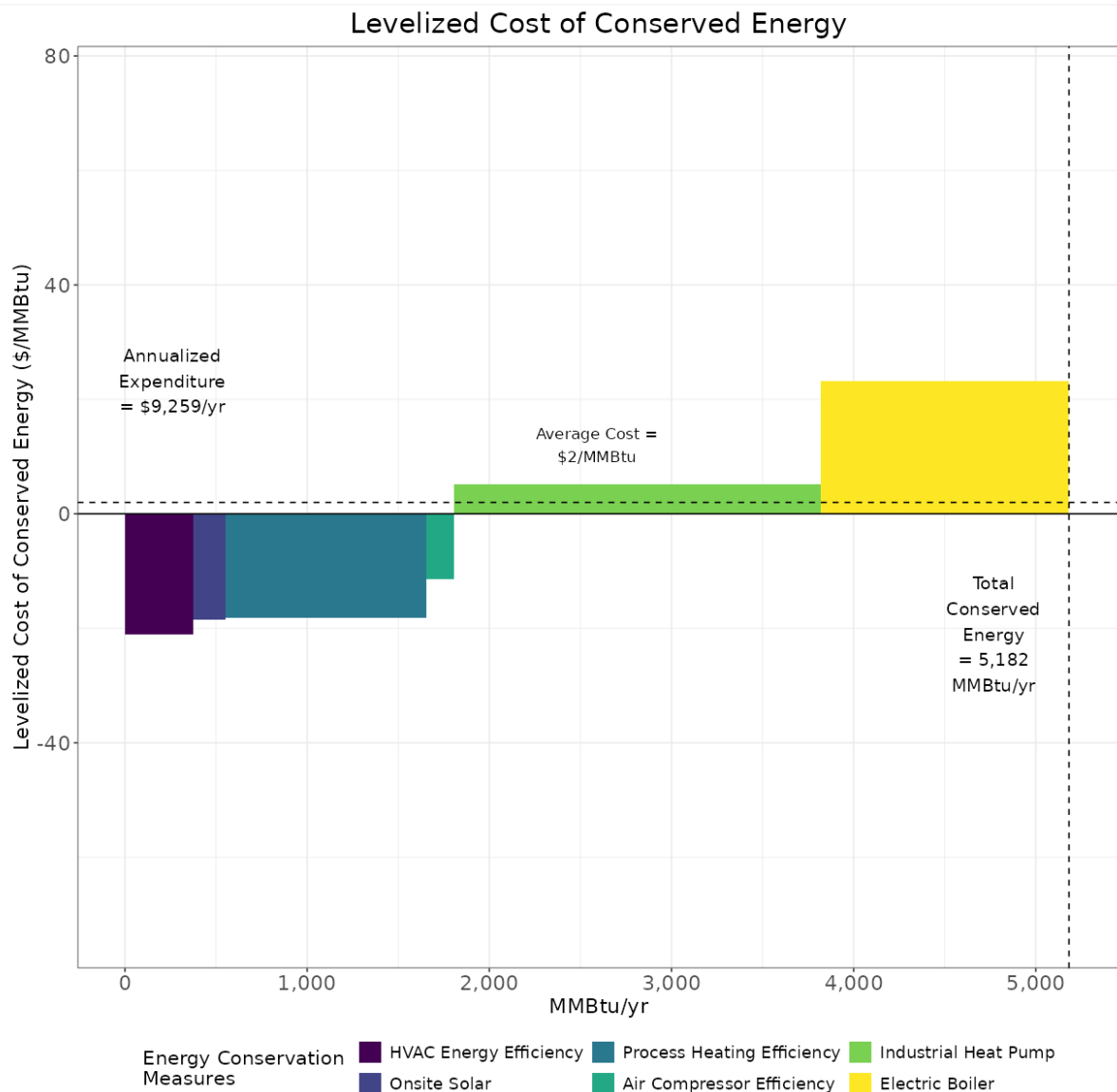


Figure 1: Example Levelized Cost Curve generated using the Levelized Cost Curve Tool

There are two components of this tool; one is an Excel-based calculator that lets users input energy and cost data to calculate the levelized costs and the other is a web component that lets users upload the populated Excel file from the calculator to create the levelized cost curves. Let's discuss levelized cost of conserved energy, the core metric of this tool, before individually exploring both components of this tool.

Levelized Cost of Conserved Energy (LCCE)

Similar in concept to the levelized cost of electricity, LCCE gives a levelized estimate of per unit cost of energy conservation based on the lifetime costs and savings associated with a recommended efficiency or conservation measure. The levelized estimate comes from the

series present worth calculations where the series present worth factor (SPWF) can be calculated from the following equation:

$$SPWF = \left(\frac{1 - (1 + i)^{-n}}{i} \right)$$

Multiplying future costs with this factor gives a levelized estimate of the present value of future costs. This is appropriate for costs that do not escalate over time like fixed operation and maintenance cost. For escalating series costs like the cost of electricity or natural gas, escalated series present worth (ESPWF) can be used.

$$ESPWF = \frac{1 - \left(\frac{1 + e}{1 + i} \right)^n}{i - e}$$

To understand how this factor enables us to estimate the present value of future costs, let's take the example of operational cost of an electric boiler. Let's assume that we have an electric boiler that costs \$5,000 to operate in the current year. Assuming that this operational cost is entirely the cost of electricity for running the electric boiler, and assuming that the electricity cost is going to annually increase at 5% (e). Let's calculate the present value of this operational cost over 10 years (n) with a discount rate (i) of 10%.

$$ESPWF = \frac{1 - \left(\frac{1 + 0.05}{1 + 0.1} \right)^{10}}{0.1 - 0.05} = 7.44$$

$$\text{Present Value of 10 year Electric Boiler Operational Cost} = 7.44 * 5,000 = \$37,199$$

By using these two series present worth factors, we can account for both fixed and escalating costs associated with the electric boiler to estimate its net costs and savings over its lifetime. This net costs and savings estimate can then be annualized by dividing it by SPWF and the resulting number can be divided by the annual conserved energy from the electric boiler to get the LCCE. It is expressed in the following equation:

$$LCCE = \left(\frac{1}{\text{Annual Conserved energy}} \right) \left(\frac{IC + FO\&M \cdot SPWF + EBOC \cdot ESPWF_{Elec} - NGCS \cdot ESPWF_{NG}}{SPWF} \right)$$

IC:	Investment cost.
FO&M:	Fixed operation and maintenance cost excluding the energy costs.
SPWF and ESPWF:	Series and escalated series present worth factors.
EBOC:	Electric boiler energy usage cost.
NGCS:	Natural gas cost savings.

Using the Excel Based Levelized Cost Calculator

The Excel based calculator allows the user to calculate the levelized cost of conserved energy by entering some facility and measure-specific inputs. The calculator has notes added

next to every input cell to help users throughout the input process. Users must navigate to one of the Input tabs in the worksheet to get started. The entries in the first input tab carry over to the second input tab for the users' convenience. There is a color-coding legend on top of this sheet to guide the users through the nature of cells (see Figure 2). The input cells are shaded light yellow and output cells are shaded light blue.

Input Cells	Fill from Left to Right
Output Cells	Filled with Calculated Outputs

Figure 2: Input and output cells

Overwriting the output cells will overwrite the formulas in those cells, rendering the sheet not suitable for a new set of inputs. If users overwrite output cells for their analysis, they must download a new copy of the input sheet for further analyses. The user starts with entering the titles of the assessment recommendations following the energy use, conservation, cost, and savings alongside the financial inputs to allow the expression presented above to calculate the LCCE for each measure (see Figure 3). After entering these inputs for all recommendations, the user saves the filled input sheet and uploads it to the web tool to generate the levelized cost curves. The input sheet has the flexibility to allow users to enter grid specifications and fuel characteristics for extended analysis beyond just energy conservation. The guided input system blackens the input cells not relevant to the energy source selection by the user. For example, if the measure is 'process heating efficiency' and it only corresponds to natural gas i.e. the user specifies natural gas as the primary energy source (without any secondary or switched energy source), the calculator blackens the input cells corresponding to electricity and fuel switching. The blackened cells are also ignored by the calculator even if they are left mistakenly populated while overwriting existing inputs.

Assessment Recommendation	Primary Energy Source	Secondary Energy Source	Switched Energy Source (Only select if switching fuels)	Annual Electricity Saved (Avoided Consumption)	Annual Primary Fuel Saved (Avoided Consumption)	Annual Secondary Fuel Saved (Avoided Consumption)	Switched Fuel Energy Consumption	Annual Switched Fuel Energy Cost	Annual Electricity Cost Savings (Avoided Cost)	Annual Fuel Cost Savings (Avoided Cost)
	If switching fuels, select the fuel to switch FROM	(Leave blank for single fuel systems)	If switching fuels, select the fuel to		Do not enter electricity saved	Do not enter electricity saved	Fuel switched TO			
				kWh/yr	MMBtu/yr	MMBtu/yr	MMBtu/yr or kWh/yr	\$/yr	\$/yr	\$/yr
Air Compressor Efficiency	Electricity			44,000					\$ 4,500	
Process Heating Efficiency	Natural Gas				1,100					\$ 16,000
HVAC Energy Efficiency	Electricity	Natural Gas		37,000		250			\$ 11,000	\$ 3,000
Industrial Heat Pump	Natural Gas		Electricity		2,600		171,000	\$ 19,000		\$ 16,800
Electric Boiler	Natural Gas		Electricity		2,000		187,000	\$ 21,000		\$ 14,500
Onsite Solar	Electricity			52,000					\$ 9,000	

Figure 3: Main input sheet

Manual Input of LCCE Values

In case the users do not want to use the built-in calculator and already have the annualized total costs and annual conserved energy values for each measure, they can bypass the calculator. These values must be entered into their corresponding output columns at the very end of the table (see Figure 4). The tool will automatically calculate the LCCE for each measure

and be ready for creating the levelized cost curve based on the names of the ARs and the calculated LCCE values. **It is important to note that if the users overwrite the default input sheet with manual input values, it overwrites the formulas written to calculate the levelized costs in the calculator. Therefore, the worksheet cannot be reused for LCCE calculation.** The users must re-download the default input sheet with preserved formulas if they overwrite the downloaded file with the manual LCCE input. The two output cells users must populate to bypass the calculator and make it visualization-ready are outlined by the red dashed line in the screenshot below:

Assessment Recommendation	Annualized Total Costs	Annual Conserved Energy - MMBtu	Levelized Cost of Conserved Energy - MMBtu	Annual Conserved Energy - MWh	Levelized Cost of Conserved Energy - MWh
	ATC	ACE	LCCE = ATC/ACE	ACE	LCCE = ATC/ACE
	$ATC = PV_{costs}/SPWF$	MMBtu/yr	\$/MMBtu	MWh/yr	\$/MWh
Air Compressor Efficiency	\$ (1,705.1)	150.1	\$ (11.4)	44	\$ (38.75)
Process Heating Efficiency	\$ (20,062.0)	1100.0	\$ (18.2)	322	\$ (62.23)
HVAC Energy Efficiency	\$ (7,905.4)	376.2	\$ (21.0)	110	\$ (71.69)
Industrial Heat Pump	\$ 10,572.5	2016.5	\$ 5.2	591	\$ 17.89
Electric Boiler	\$ 31,628.3	1361.9	\$ 23.2	399	\$ 79.24
Onsite Solar	\$ (3,269.2)	177.4	\$ (18.4)	52	\$ (62.87)

Figure 4: Economic calculations in the main input sheet

Inputs for Assessment Recommendations

This section goes over all these inputs and provides users with some context about what each input is for with examples based on a demonstrative assessment entered into the tool. Note that if you hover over the input column title, you'll see a popup note to give a short-version of the detailed account given here (see Figure 5).

Assessment Recommendation	Primary Energy Source	Select the energy source that the respective assessment recommendation primarily uses. For example, you'd select Natural Gas from the dropdown if you are writing the natural gas boiler efficiency assessment recommendation. This cell can be confusing for recommending fuel switching like for instance in case of heat pumps. In that instance you'd enter the energy source being switched FROM i.e. for switching from natural gas boilers to heat pumps, you'd select Natural Gas as the primary energy source.
	If switching fuels, select the fuel to switch FROM	
Air Compressor Efficiency	Electricity	
Process Heating Efficiency	Natural Gas	
HVAC Energy Efficiency	Electricity	
Industrial Heat Pump	Natural Gas	
Electric Boiler	Natural Gas	
Onsite Solar	Electricity	

Figure 5: Energy sources

Title of Assessment Recommendation

First input is the title of the assessment recommendation as the users want shown on the levelized cost curve. It is important that users avoid using the same name for two recommendations to avoid visualization glitches on the web-version of the tool. For example, if a user splits the HVAC efficiency recommendation into electricity and natural gas savings, we recommend hyphenating it as HVAC-Electricity or HVAC-Gas to avoid confusion and glitching (see Figure 6).

Assessment Recommendation	Primary Energy Source
	If switching fuels, select the fuel to switch FROM
Air Compressor Efficiency	Electricity
Process Heating Efficiency	Natural Gas
HVAC Energy Efficiency	Electricity
Industrial Heat Pump	Natural Gas
Electric Boiler	Natural Gas
Onsite Solar	Natural Gas

Figure 6: Assessment recommendations and primary energy sources

Energy Sources for Assessment Recommendations

Next three inputs ask users to specify the energy source corresponding to the assessment recommendation. All of these inputs are to be **selected by the users from the drop-down list of energy sources**. The list may not include any atypical energy sources, but users can enter their specifications in the last tab of the worksheet.

You can see in the figure below that there can be up to two energy sources per assessment recommendation. For example, for the demonstrative assessment report, the first three recommendations are efficiency related. If the team performing the assessment wanted to combine all these measures (thermal and electrical) at the facility as one recommendation, the energy sources would be both electricity and natural gas. Therefore, there is an option to assign two energy sources to the same measure. This is exemplified in the HVAC Energy Efficiency measure in the example below, where primary energy source is electricity and secondary energy source is natural gas. Other measures are the examples of single-source recommendations.

The next input column is for the recommendations involving fuel switching. For example, switching from natural gas to biogas or electricity is a fuel switching recommendation. For the demonstration, we have two such recommendations, 'Industrial Heat Pump' and 'Electric Boiler'. **Users must specify the fuel that will be used AFTER fuel switching in this column**, which is electricity for both recommendations. **Notice how 'Primary Energy Source' column has the fuel being switched FROM and the 'Switched Energy Source' column has the fuel being switched TO (see Figure 7).** This column is only applicable to the recommendations involving fuel switching and is blank for other recommendations as shown below.

Assessment Recommendation	Primary Energy Source	Secondary Energy Source	Switched Energy Source (Only select if switching fuels)
	If switching fuels, select the fuel to switch FROM	(Leave blank for single fuel systems)	If switching fuels, select the fuel to
Air Compressor Efficiency	Electricity		
Process Heating Efficiency	Natural Gas		
HVAC Energy Efficiency	Electricity	Natural Gas	
Industrial Heat Pump	Natural Gas		Electricity
Electric Boiler	Natural Gas		Electricity
Onsite Solar	Electricity		

Figure 7: Secondary and switched energy sources

Energy Savings

The next three inputs require users to specify the energy savings in MMBtu or kWh for the assessment recommendations. These savings come from the avoided consumption of energy resulting from a recommendation. In the demonstrative assessment for example, the energy efficiency recommendation results in the avoided consumption of both electricity and natural gas. **Notice that there is a dedicated input column for the avoided consumption of electricity titled 'Annual Electricity Saved' so the users must not enter electricity savings in the 'Annual Primary Non-Electricity Fuel Saved' column.** This is because the calculator is tuned to look up the grid characteristics separately from the fossil fuel characteristics.

When entering the energy savings (avoided consumption) in these columns, the users must ensure that the units are consistent with the given units in the column description. These units for electricity savings are kWh/year and for non-electricity energy sources, they are MMBtu/year. Not doing so would result in inaccurate estimates of levelized costs. This is because the grid electricity and fuel characteristics are respectively denominated by these units (see Figure 8).

Assessment Recommendation	Annual Electricity Saved (Avoided Consumption)	Annual Primary Fuel Saved (Avoided Consumption)	Annual Secondary Fuel Saved (Avoided Consumption)
		Do not enter electricity saved	Do not enter electricity saved
	kWh/yr	MMBtu/yr	MMBtu/yr
Air Compressor Efficiency	44,000		
Process Heating Efficiency		1,100	
HVAC Energy Efficiency	37,000		250
Industrial Heat Pump		2,600	
Electric Boiler		2,000	
Onsite Solar	52,000		

Figure 8: Energy savings columns

Impact from Fuel Switching

Next two input columns incorporate the additional energy consumption and the associated energy cost with fuel switching into the levelized cost calculation. The first of these two columns ask the users to enter the additional energy consumption coming from fuel switching. For example, in case of heat pumps in the demonstrative assessment, we can see that an additional 171,000 kWh/year is consumed because of this recommendation. For electric boilers, additional 187,000 kWh/year is consumed. The calculator assigns the unit MMBtu/yr or kWh/yr to the entered values in this column based on the drop-down list

selection made by the user in the column titled 'Switched Energy Source' (Figure 9). The next column asks the users to enter the additional cost associated with fuel switching. This is essentially the energy cost for the additional kilowatt-hours of electricity or MMBtu of non-electricity fuel. For the demonstrative assessment, the additional 171,000 kWh/year from heat pumps and 187,000 kWh/year from electric boilers cost \$19,000 and \$21,000 respectively.

Assessment Recommendation	Switched Fuel Energy Consumption	Annual Switched Fuel Energy Cost
	Fuel switched TO	
	MMBtu/yr or kWh/yr	\$/yr
Air Compressor Efficiency		
Process Heating Efficiency		
HVAC Energy Efficiency		
Industrial Heat Pump		
Electric Boiler	171,000	\$ 19,000
Onsite Solar	187,000	\$ 21,000

Figure 9: Switched energy costs

Energy Cost Savings

Next two input columns ask users to enter the cost savings resulting from the assessment recommendations. These inputs are split into two columns with the first one asking for the annual electricity cost savings and the second one asking for the annual non-electricity fuel cost savings (see Figure 10). **The users must incorporate utility and facility specific demand charges or time-of-use tariffs (if applicable) to calculate these savings for both electricity and non-electricity fuels.** For non-electricity fuel cost savings, users may have to add primary and secondary fuel cost savings for measures corresponding to operations with multiple fuels. **Users may enter 0 or leave the cell empty for recommendations that do not offer either electricity or non-electricity fuel cost savings.** This is shown in the demonstrative assessment for Process Integration, Industrial Heat Pumps and Electric Boilers having zero electricity savings. Notice how the tool has blackened these cells as they do not apply to the energy source profiles of these measures.

Assessment Recommendation	Annual Electricity Cost Savings (Avoided Cost)	Annual Fuel Cost Savings (Avoided Cost)
	\$/yr	\$/yr
Air Compressor Efficiency	\$ 4,500	
Process Heating Efficiency		\$ 16,000
HVAC Energy Efficiency	\$ 11,000	\$ 3,000
Industrial Heat Pump		\$ 16,800
Electric Boiler		\$ 14,500
Onsite Solar	\$ 9,000	

Figure 10: Annual costs savings related to energy

Financial Inputs

Next four input columns ask users to enter some important financial inputs for calculating the levelized costs from the assessment recommendations. First one, titled **'Investment Cost'**, asks the users to enter the total capital cost, including all other upfront costs like installation and retrofit costs (if any) for the recommended measures (see Figure 11). For example, for Heat Pumps, the investment cost of \$39,000 includes the costs of heat pumps, hot water pipe infrastructure and the accompanying thermal storage tank etc. along with their installation costs. The next input column asks the users to enter the fixed operation and maintenance cost associated with the equipment (if any). These costs may include but not be limited to the scheduled maintenance or servicing of the equipment.

Next two inputs are critical to the financial analysis done by this calculator. As we saw in the calculation of series and escalated series present worth factors in the introductory section about the levelized cost of conserved energy, these two parameters greatly influence the results. Discount rate is used to translate the future costs into present value. It reflects how much more you value your money today versus in the future, a lower rate favoring the future and higher rate tilting toward the present. Project lifetime is the number of years we expect the equipment to be functional or keep giving us the costs and savings that we entered in the input columns before. **The default values of 5% for the discount rate and 10 and 20 years for the project lifetime are entered into the calculator that must be edited by the user based on the assessment recommendation and the facility's preferences.**

Assessment Recommendation	Implementation Cost	Annual Fixed Operation and Maintenance Cost <i>(optional)</i>	Discount Rate	Project Lifetime
	IC	O&M	i	n
	\$	\$/yr		
Air Compressor Efficiency	\$ 17,000	\$ 1,300	5%	10
Process Heating Efficiency	\$ 8,000	\$ 700	5%	20
HVAC Energy Efficiency	\$ 95,000	\$ 3,200	5%	20
Industrial Heat Pump	\$ 39,000	\$ 4,500	5%	20
Electric Boiler	\$ 120,000	\$ 8,567	5%	10
Onsite Solar	\$ 32,000	\$ 3,000	5%	10

Figure 11: Financial input variables

Energy Cost Escalation Rates

Finally, the users can enter the escalation rates for the different costs going into the calculation of levelized costs (see Figure 12). There are two input columns corresponding to the electricity and non-electricity fuel cost escalation rates. For convenience, default escalation rates of 3.5% are added for electricity and non-electricity fuel cost escalation. **Users may enter a more appropriate energy cost escalation rate or may consult with the facility to ask them about their preference regarding this.**

Assessment Recommendation	Electricity Cost Escalation Rate	Fuel Cost Escalation Rate
	e1	e2
Air Compressor Efficiency	3.5%	
Process Heating Efficiency		3.5%
HVAC Energy Efficiency	3.5%	3.5%
Industrial Heat Pump	3.5%	3.5%
Electric Boiler	3.5%	3.5%
Onsite Solar	3.5%	

Figure 12: Energy annual escalation rates

Generating File for Online Visualization Tool

After all these inputs, the users can see the LCCE in the right most column of the main inputs table. This column is titled 'Levelized Cost of Conserved Energy' and is shaded blue as it is the output cell. The LCCE is given in two columns with one reporting it in \$/MMBtu and the other reporting it in \$/MWh. The users must save the changes made to the input sheet before uploading it to the visualization component of the tool.

Visualization Component of the Tool

The visualization component of the tool is hosted online at the Lawrence Berkeley National Laboratory's server and can be accessed at <https://industrialdecARB.lbl.gov/apps/lcc/>. The screen that users will see on the web address for this tool is shown in Figure 13.

Levelized Cost Curve Tool

Load Inputs Energy Emissions

Download LCC - Input Sheet
Enter Facility Name

Upload 'LC - Input Sheet' Excel File
Browse... No file selected

Download Tool Documentation

Instructions

Tab 1: Load Inputs

1. Download and fill out Input Sheet.
2. Upload completed Input Sheet.
3. Refer to documentation if needed.

Tab 2: Energy

1. View and customize Levelized Cost of Conserved Energy (LCCE) Diagram.
2. Download LCCE Diagram.

Tab 3: Emissions

1. View and customize Levelized Cost of Avoided CO₂e (LCAC) Diagram.
2. Download LCAC Diagram.

Suggested Citation

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Figure 13: Website landing page with instructions and upload panel

The users can download the excel based calculator using the hyperlink titled 'Download LCC – Input Sheet' and download this user guide using the 'Download Tool Documentation' hyperlink. Users can enter the facility name in the field titled 'Enter Facility Name' to be shown on the levelized cost curve (LCC) generated by the tool. The 'Upload LC – Input Sheet' field lets users browse and navigate to where the input sheet has been saved to

upload it to generate the LCC. Once the user uploads the input sheet, the tool navigates to the 'Energy' tab where the users can see the LCC and adjust the font, labels, legend, and axis scaling (see Figure 14).

Levelized Cost Curve Tool

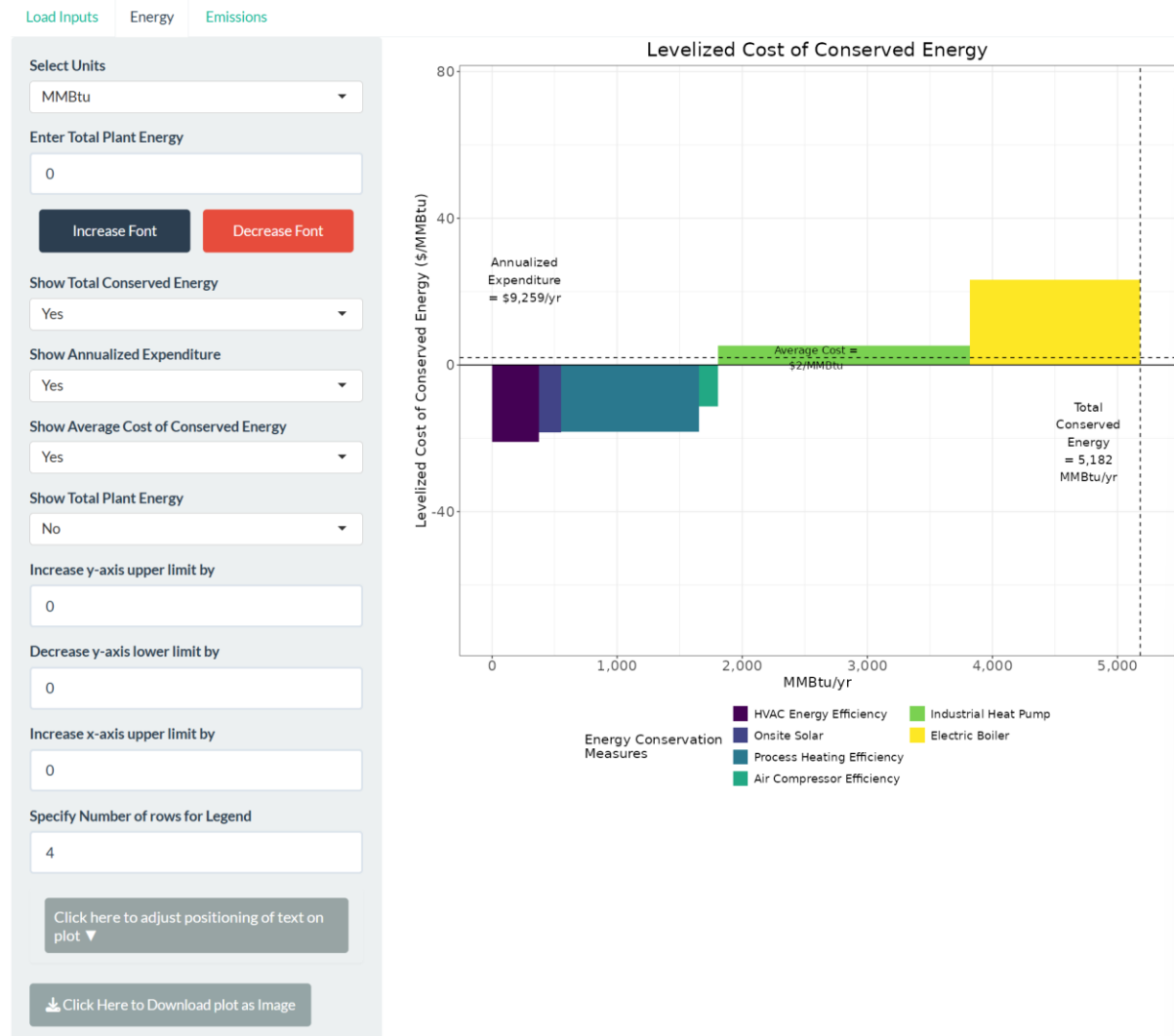


Figure 14: LCC result page

Feedback

For queries and feedback:

<https://industrialapplications.lbl.gov/industrial-assessment-guidance>